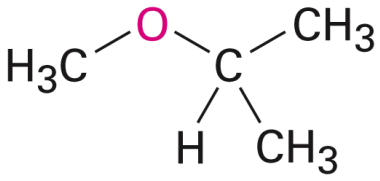
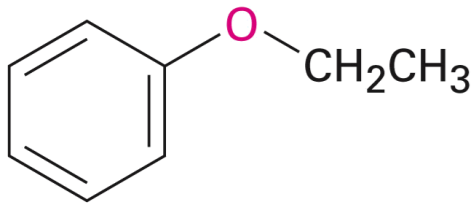
Chapter 18
Ethers and Epoxides; Thiols and Sulfides

Names and Properties of Ethers

- Simple ethers are named by identifying two organic substituents and adding the word ether

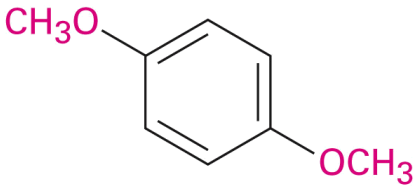


Isopropyl methyl ether

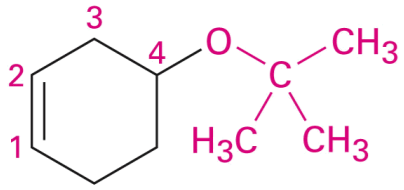


Ethyl phenyl ether

- If other functional groups are present, the ether part is considered an alk-oxy substituent



***p*-Dimethoxybenzene**



4-*tert*-Butoxy-1-cyclohexene

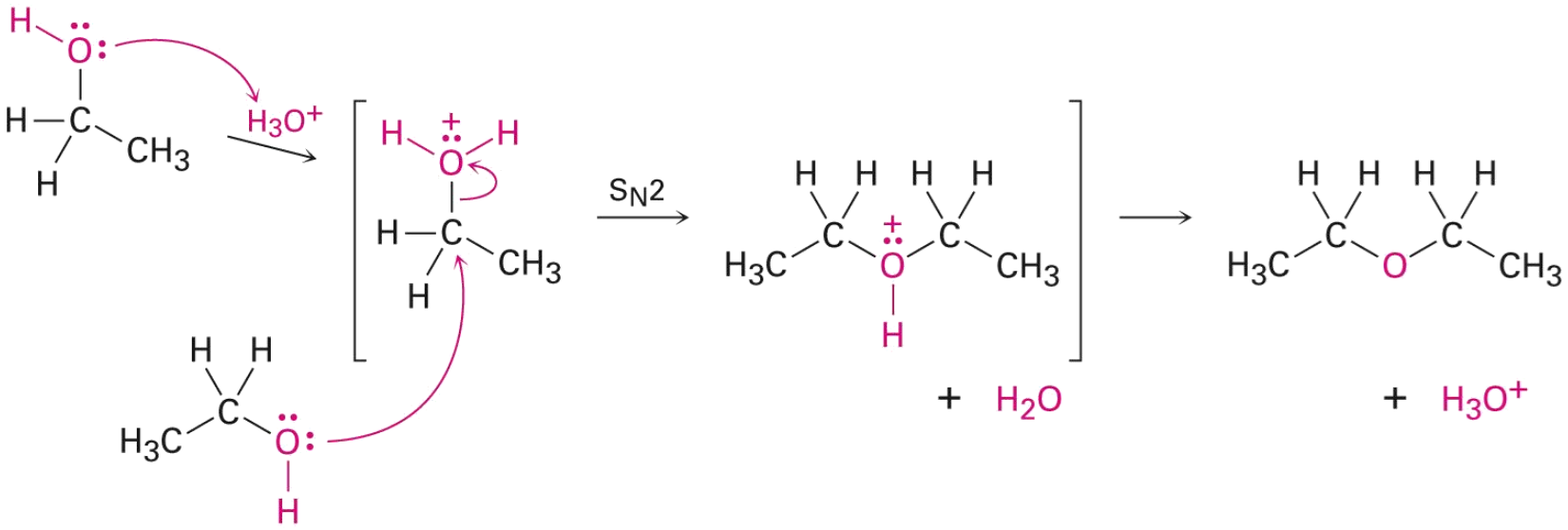
Names and Properties of Ethers

London dispersion force / hydrogen-bonding interactions

Carbon	Compound	Type	b.p. (°C)
C_2	CH_3CH_3	Hydrocarbon	−89
	CH_3OCH_3	Ether	−25
	CH_3CH_2OH	Alcohol	78
	CH_3CO_2H	Carboxylic acid	118
C_4	$CH_3CH_2CH_2CH_3$	Hydrocarbon	−0.5
	$CH_3CH_2OCH_2CH_3$	Ether	34.6
	$CH_3CH_2CH_2CH_2OH$	Alcohol	118
	$CH_3CH_2CH_2CO_2H$	Carboxylic acid	164

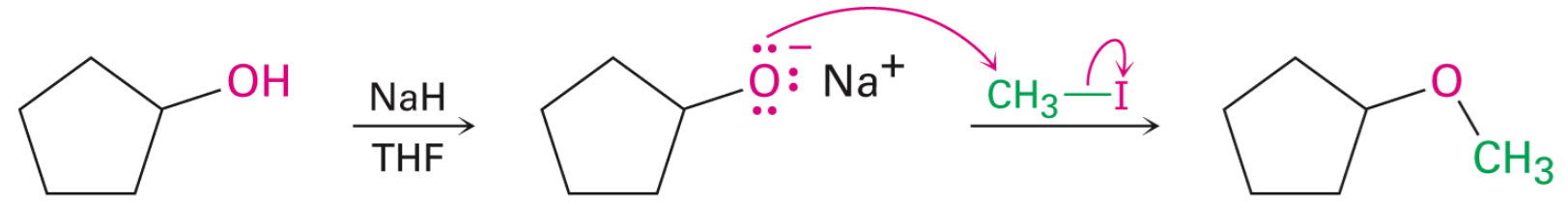
Synthesis of Ethers

- Prepared industrially by sulfuric-acid-catalyzed reaction of alcohols
  - Limited to use with primary alcohols



Williamson Ether Synthesis

- Reaction of metal alkoxides and primary alkyl halides and tosylates in an S_N2 reaction
- Best method for the preparation of ethers

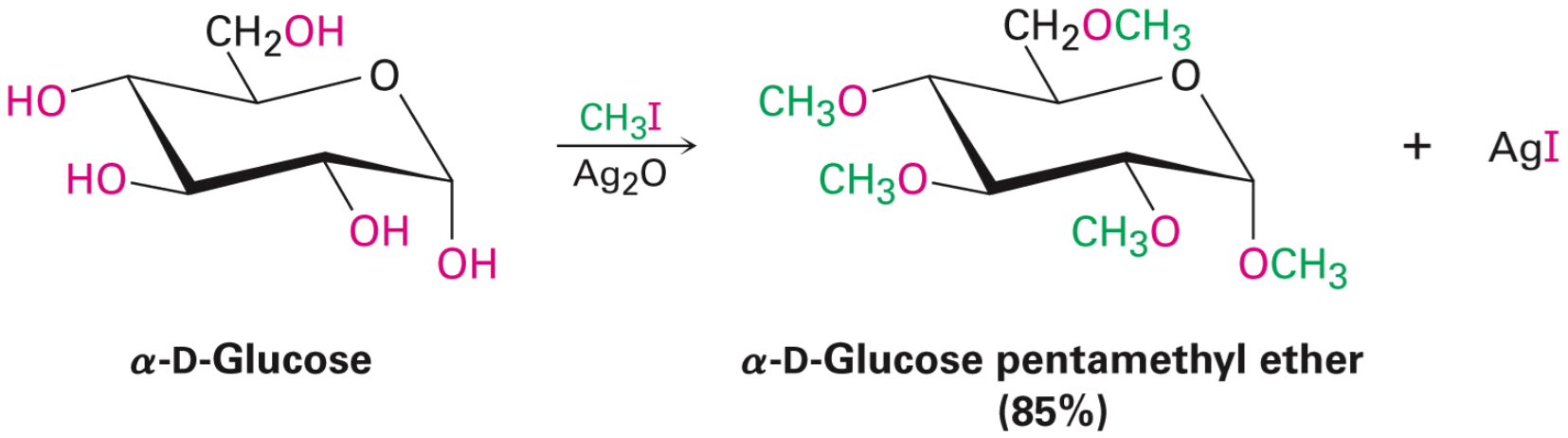


Cyclopentanol **Alkoxide ion** **Cyclopentyl methyl ether (74%)**

- Alkoxides are prepared by reaction of an alcohol with a strong base such as sodium hydride, NaH

Silver Oxide-Catalyzed Ether Formation

- Reaction of alcohols with Ag_2O directly with alkyl halide forms ether in one step
- Glucose reacts with excess iodomethane in the presence of Ag_2O to generate a pentaether in 85% yield

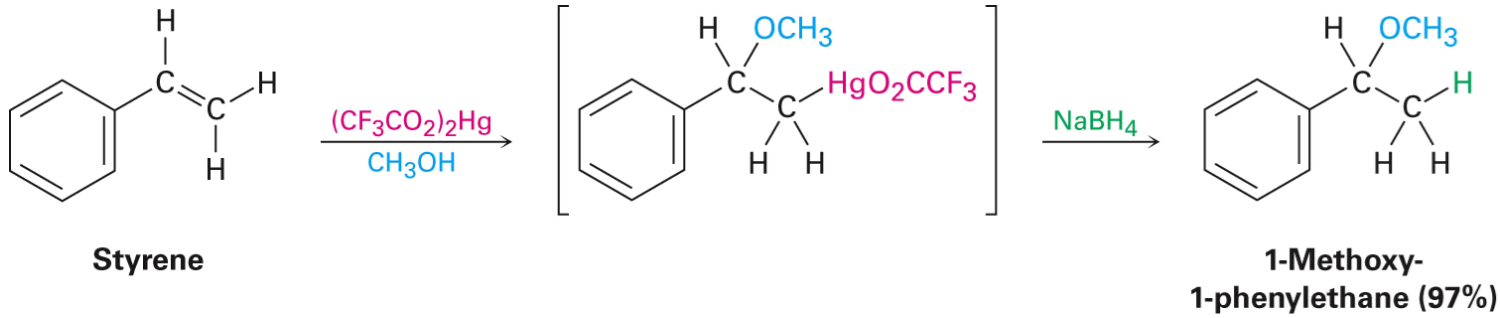


α-D-Glucose

α-D-Glucose pentamethyl ether (85%)

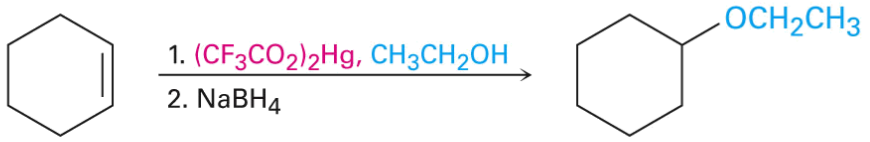
Alkoxymercuration of Alkenes

- Alkene is treated with an alcohol in the presence of mercuric acetate or trifluoroacetate
  - Demercuration with $NaBH_4$ yields an ether
- Overall Markovnikov addition of alcohol to alkene



Styrene

1-Methoxy-1-phenylethane (97%)

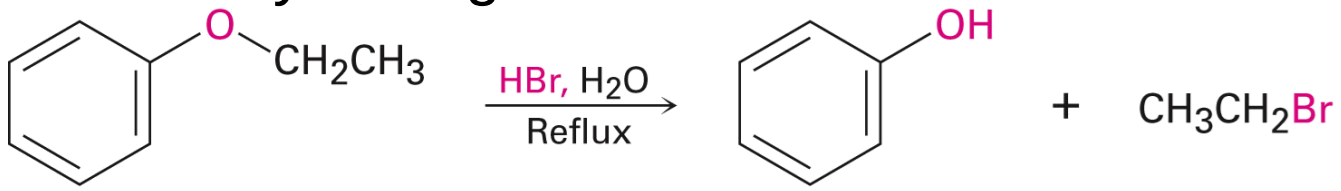


Cyclohexene

Cyclohexyl ethyl ether (100%)

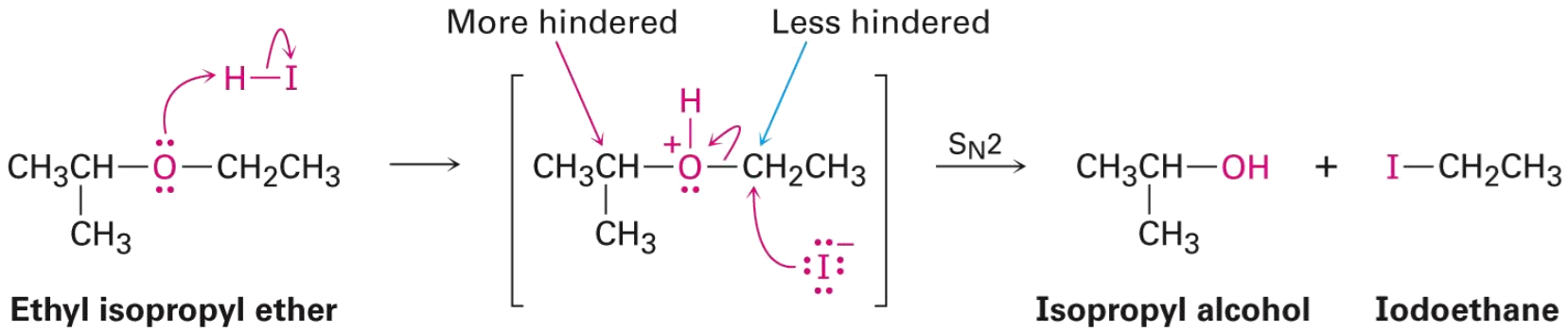
Reactions of Ethers: Acidic Cleavage

- Cleaved by strong acids



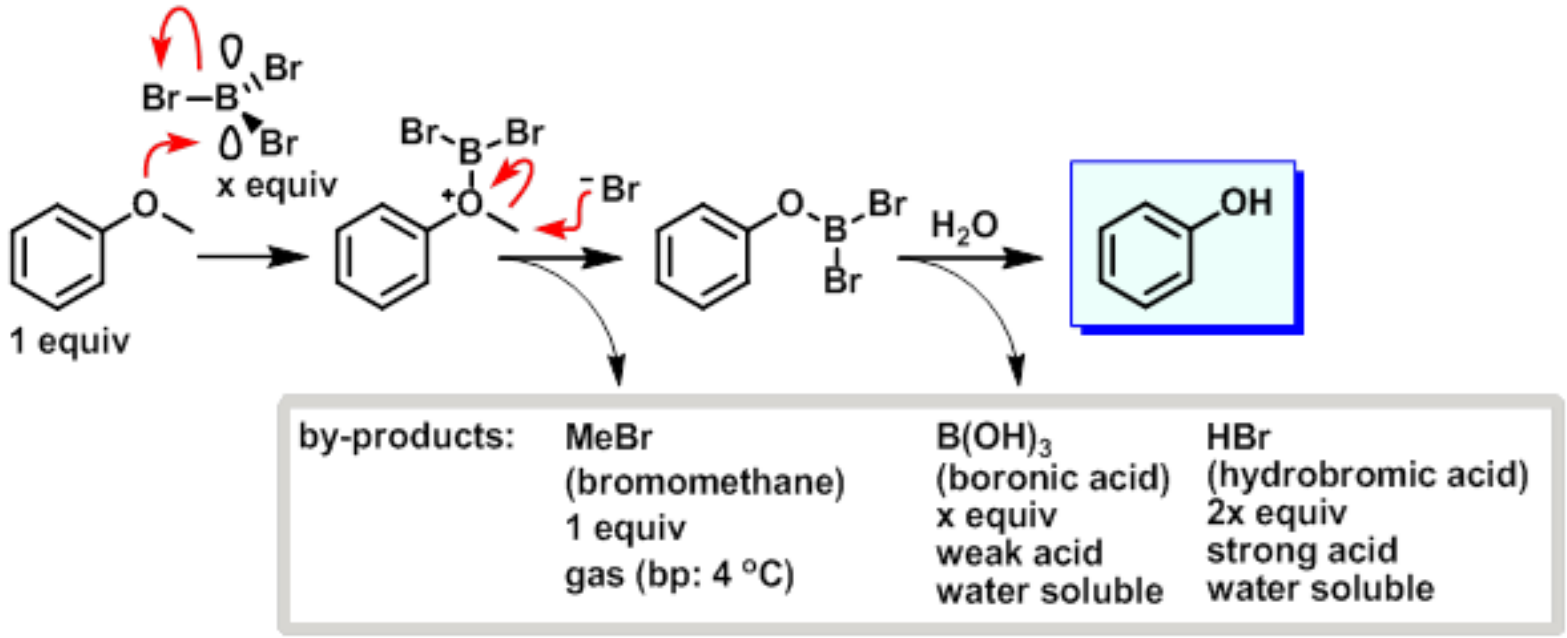
Ethyl phenyl ether **Phenol** **Bromoethane**

- HI, HBr produce an alkyl halide from less hindered component by S_N2



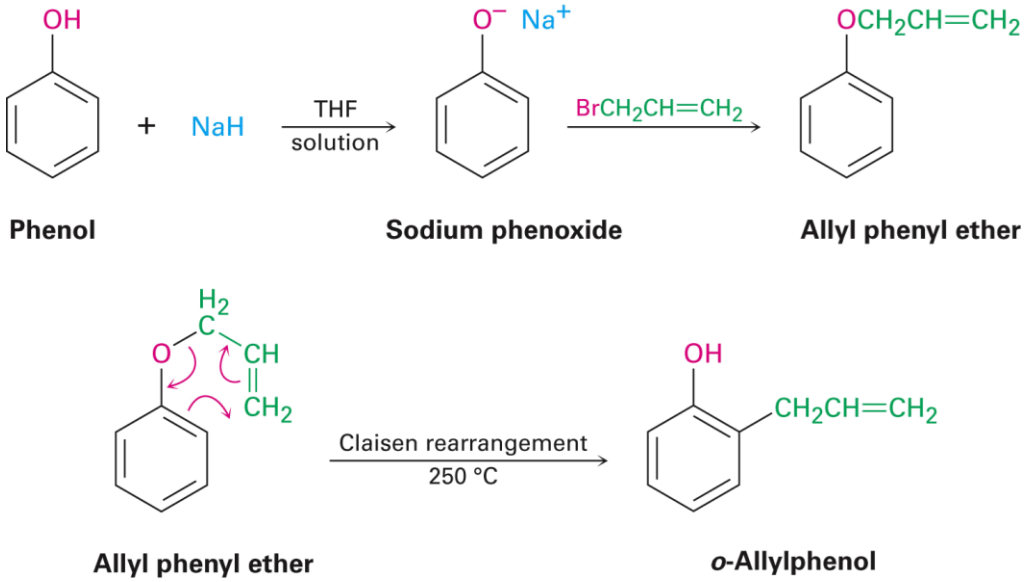
Ethyl isopropyl ether **Isopropyl alcohol** **Iodoethane**

O-Demethylation (BBr_3)



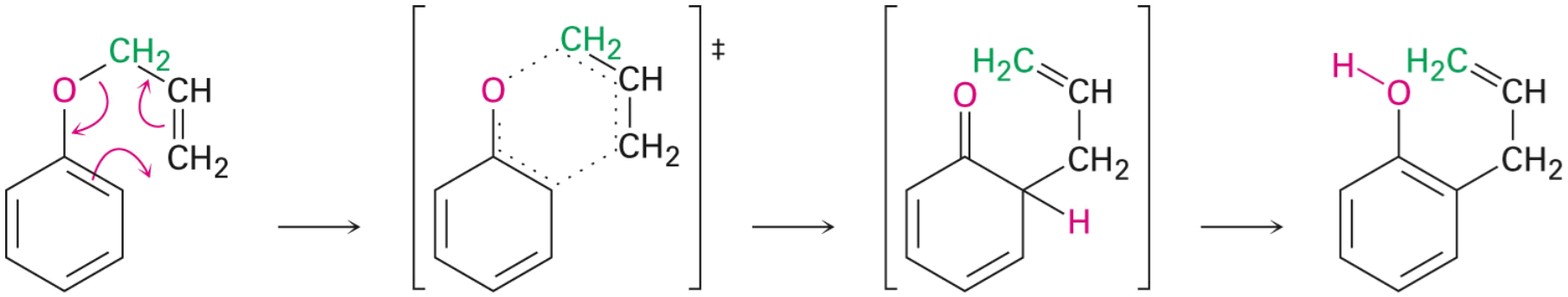
Reactions of Ethers: Claisen Rearrangement

- Specific to allyl aryl ethers and allyl vinyl ethers
- Caused by heating ally aryl ether to 200-250 ˚C
  - Leads to an o-allylphenol
- Result is alkylation of the phenol in an ortho position



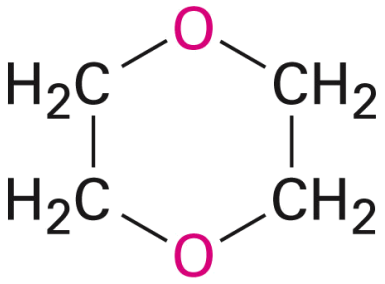
Reactions of Ethers: Claisen Rearrangement

- Takes place in a single step through a pericyclic mechanism
  - Reorganization of bonding electrons of a six-membered, cyclic transition state
- Mechanism is consistent with ^{14}C labeling

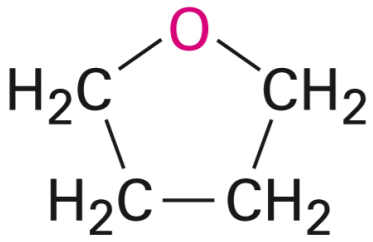


Cyclic Ethers

- Behave like acyclic ethers with the exception of three-membered ring called epoxides
  - Strain of the three-membered ring gives epoxides a unique chemical reactivity



1,4-Dioxane

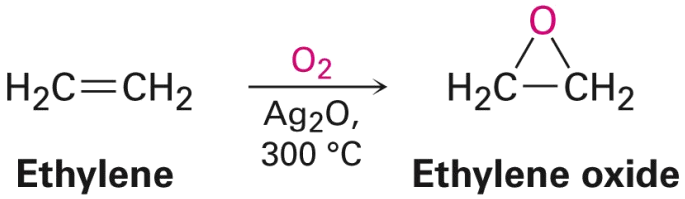


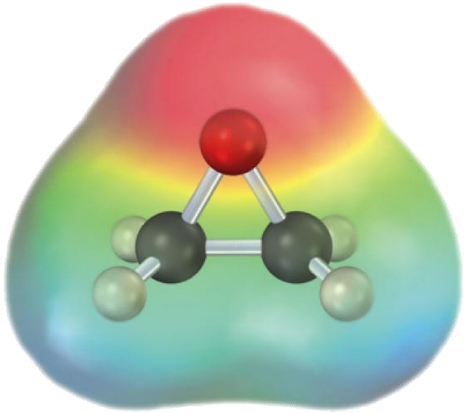
Tetrahydrofuran

- Dioxane and tetrahydrofuran are used as solvents

Cyclic Ethers

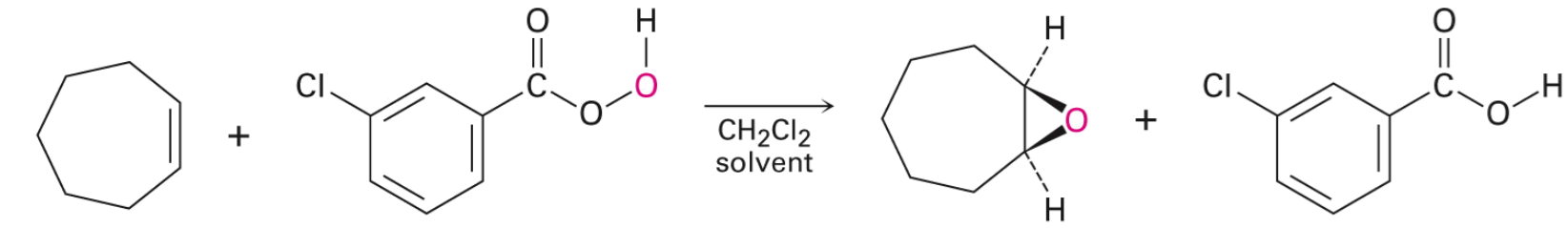
- Also called epoxides
- Ethylene oxide is industrially important as an intermediate
  - Prepared by reaction of ethylene with oxygen at 300 ° C over a silver oxide catalyst
  - -ene ending implies the presence of a double bond in the molecule





Preparation of Epoxides

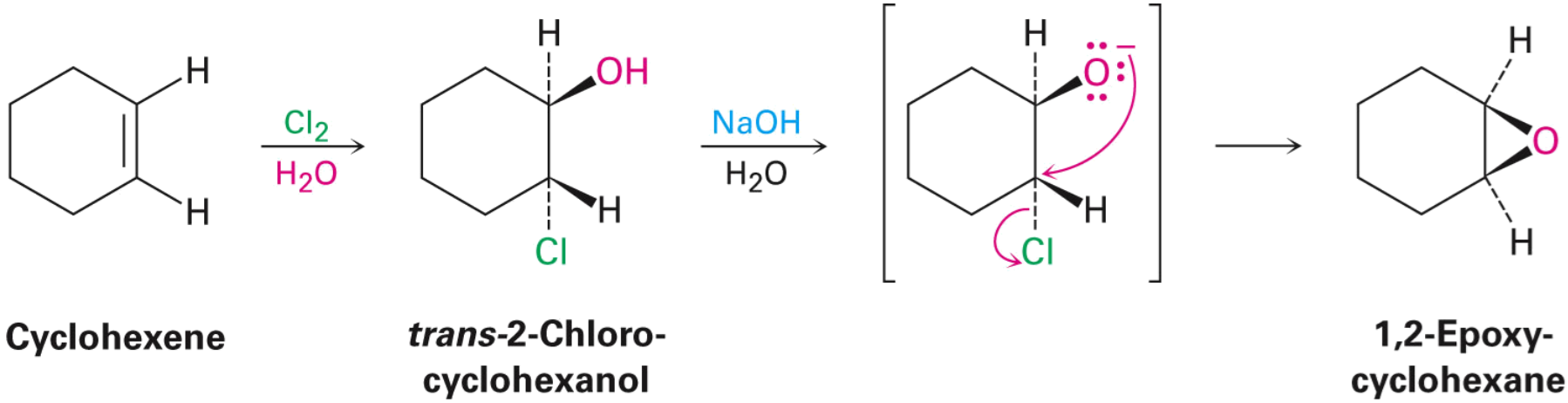
- By treating alkenes with a peroxyacid (RCO_3H)
- Also prepared from halohydrins



Cycloheptene + ***meta*-Chloroperoxy-benzoic acid** → **1,2-Epoxy-cycloheptane** + ***meta*-Chloro-benzoic acid**

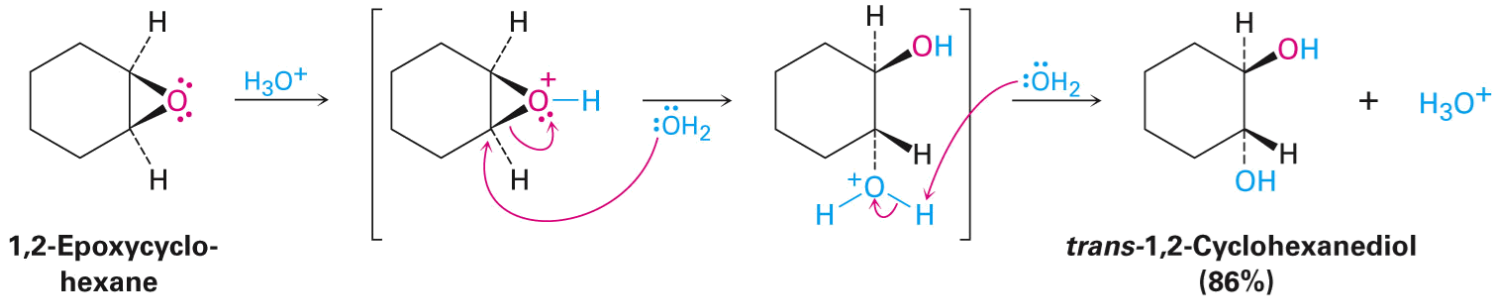
Epoxides from Halohydrins

- Addition of HO–X to an alkene gives a halohydrin
- Treatment of a halohydrin with base gives an epoxide
  - Intramolecular Williamson ether synthesis



Reactions of Epoxides: Ring-Opening (Acid cat.)

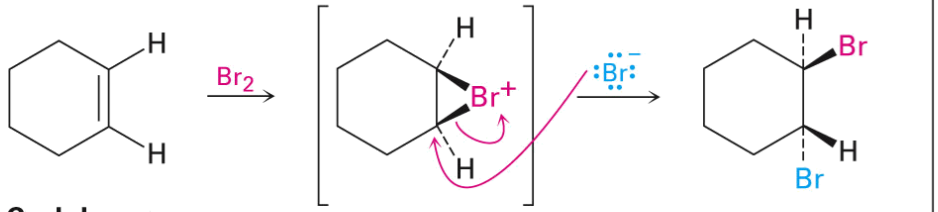
- Water adds to epoxides with dilute acid at room temperature
  - Product is a 1,2-diol



1,2-Epoxycyclohexane

trans-1,2-Cyclohexanediol (86%)

Recall the following:

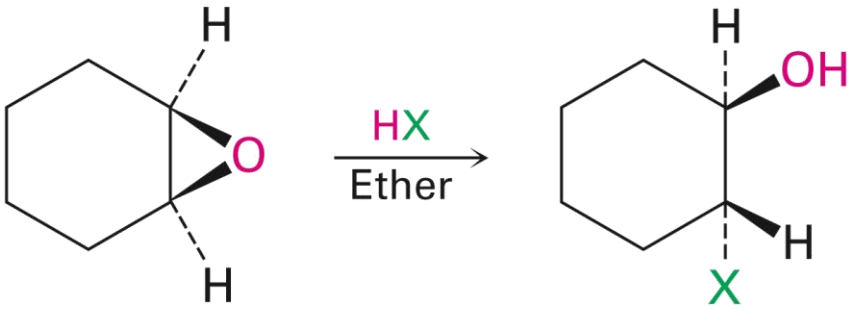


Cyclohexene

trans-1,2-Dibromocyclohexane

Reactions of Epoxides: Ring-Opening

- Also can be opened by reaction with acids other than H_3O^+
- Anhydrous HF, HBr, HCl, or HI combine with an epoxide
  - Gives a trans product



A trans 2-halocyclohexanol

where X = F, Br, Cl, or I

Reactions of Epoxides: Ring-Opening

- Regiochemistry of acid-catalyzed ring-opening depends on the epoxide's structure
- Nucleophilic attack occurs primarily at the more highly substituted site, when one epoxide carbon atoms is tertiary

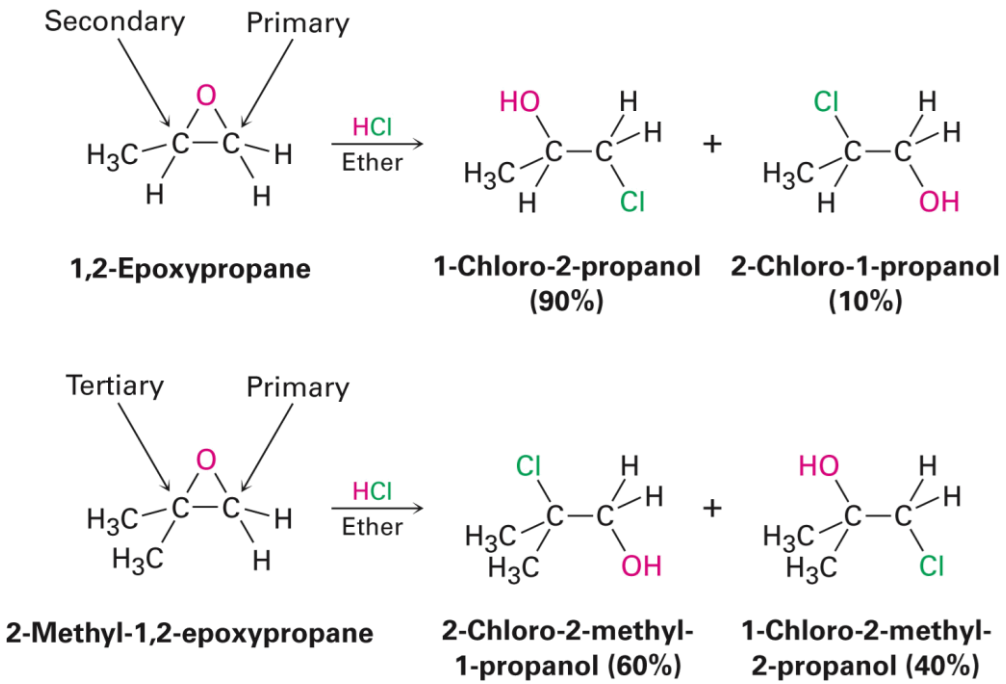
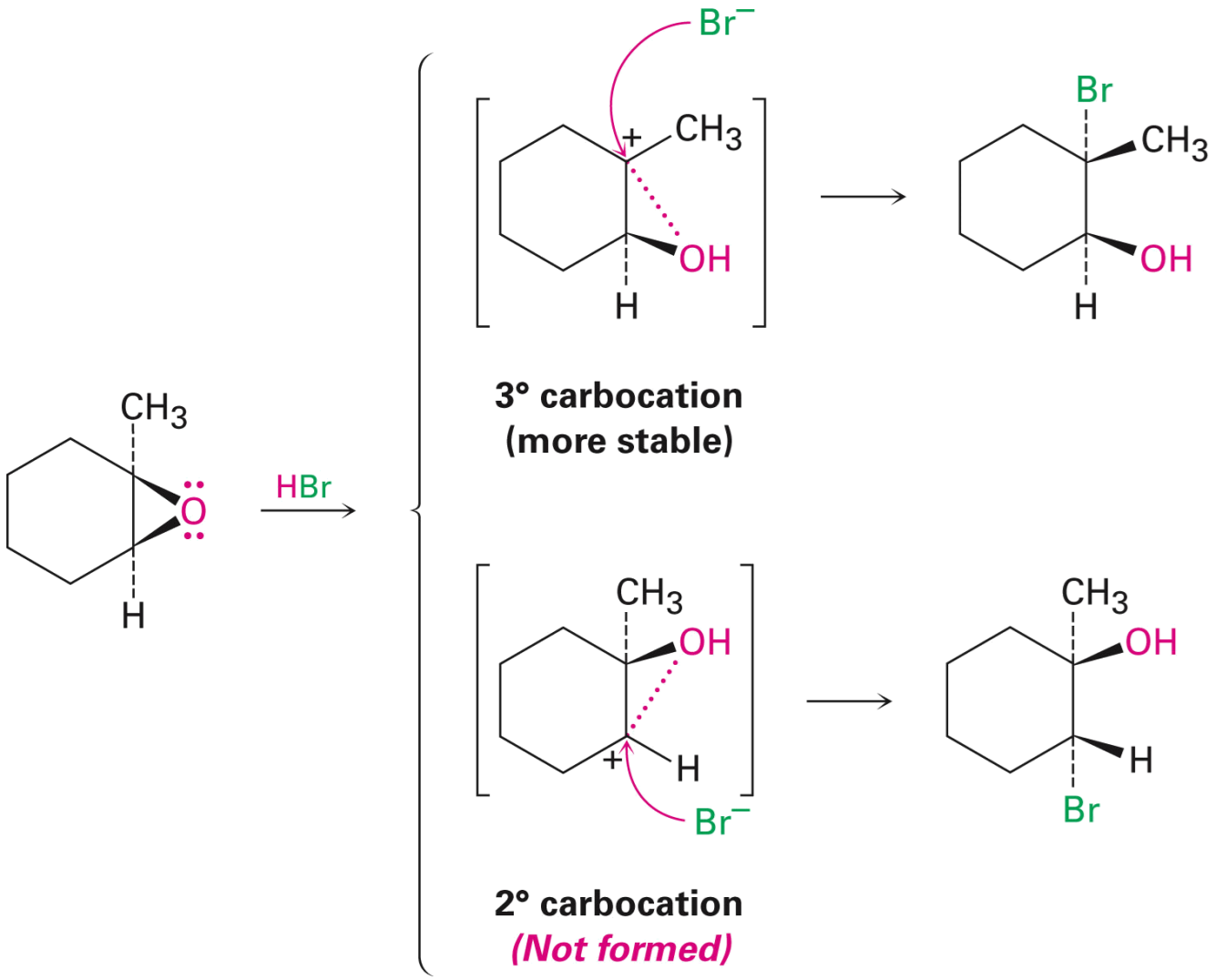
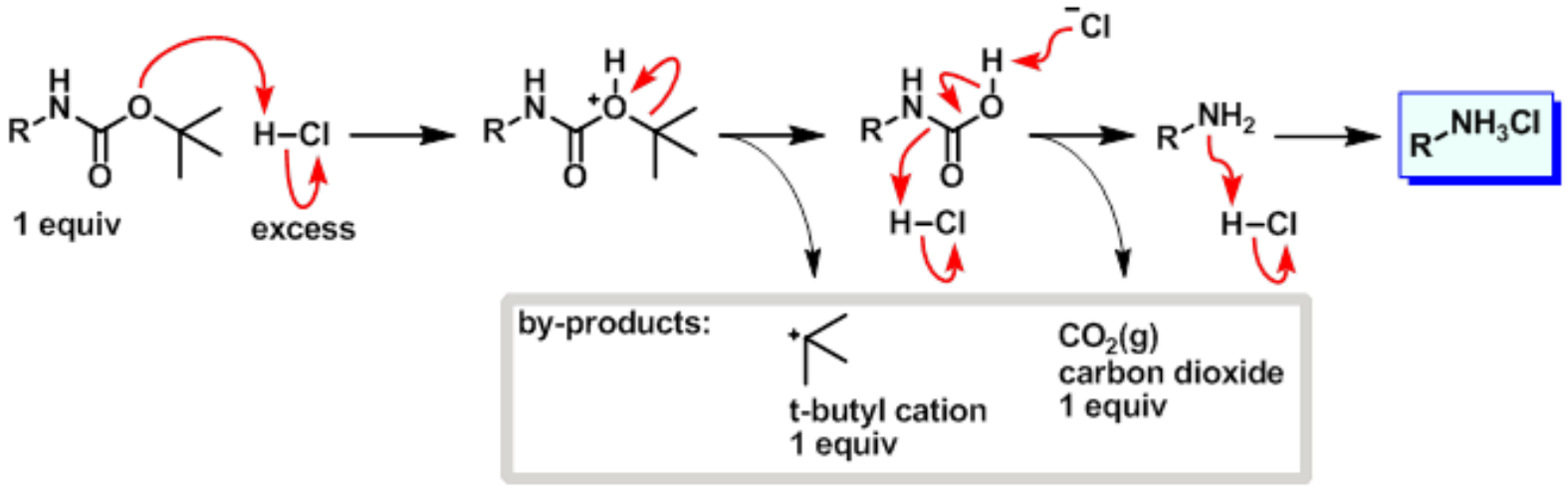


Figure 18.2 - Ring-Opening of 1,2-epoxy-1-methylcyclohexane with HBr



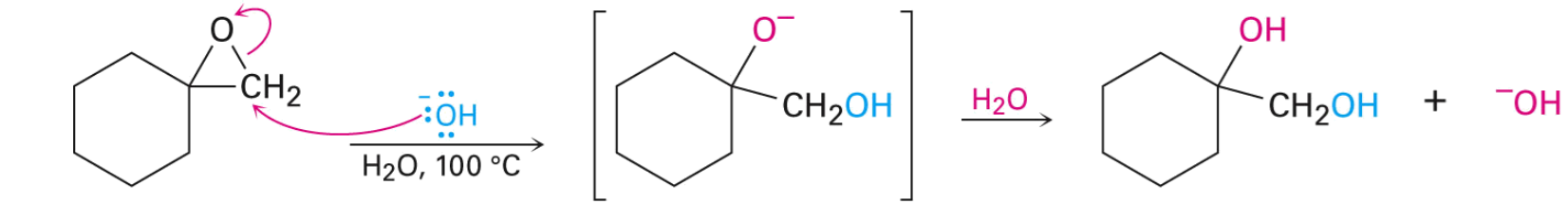
Boc (tert-butoxycarbonyl) deprotection



Organic Chemistry Portal

Base-Catalyzed Epoxide Opening

- Epoxide rings can be cleaved by bases, nucleophiles, and acids
  - Strain of the three-membered ring is relieved on ring-opening
  - Hydroxide cleaves epoxides at elevated temperatures

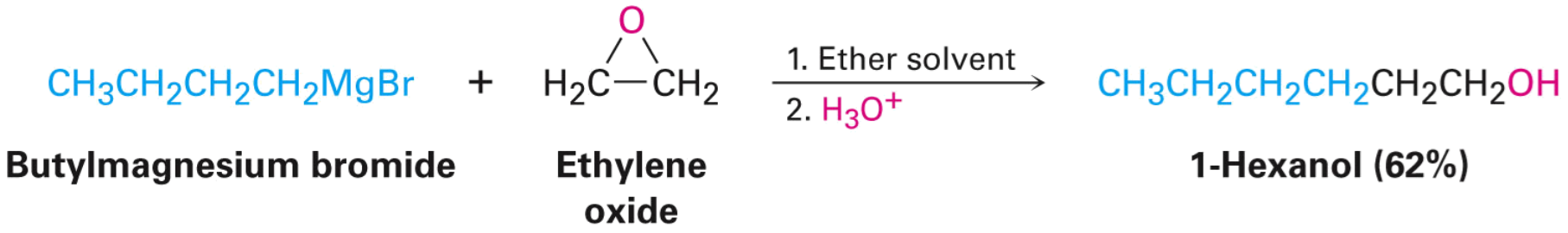


Methylenecyclohexane oxide

1-Hydroxymethyl-cyclohexanol (70%)

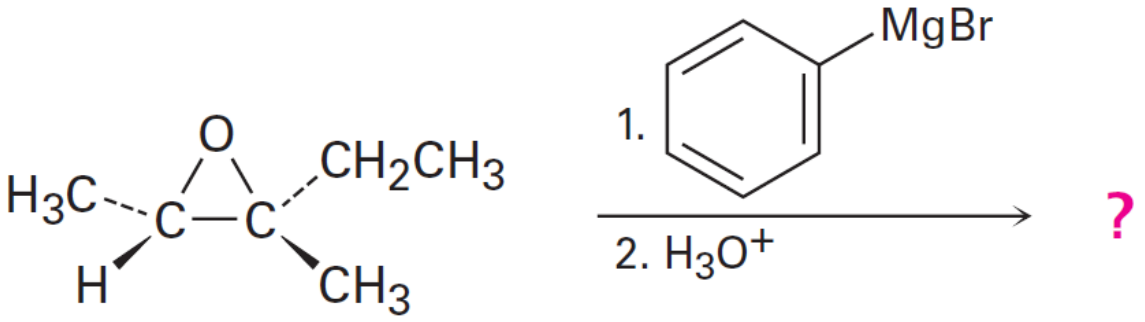
Base-Catalyzed Epoxide Opening

- Amines and Grignard reagents can be used for epoxide opening
- Ethylene oxide is frequently used
  - Allows conversion of a Grignard reagent into a primary alcohol



Worked Example

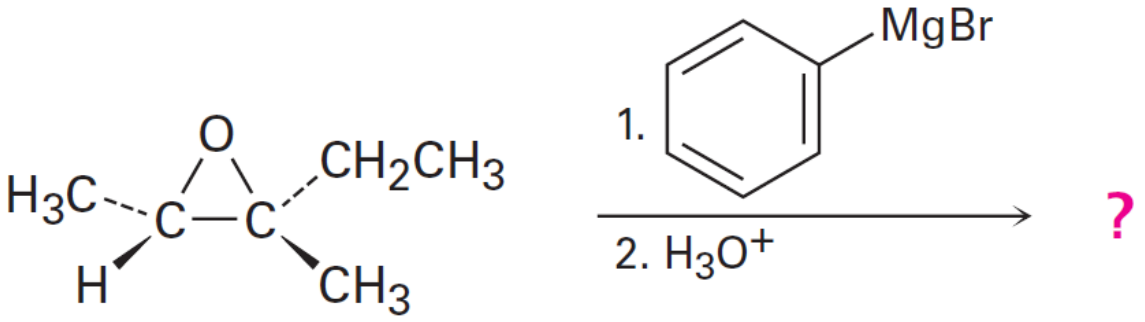
- Predict the major product of the following reaction:



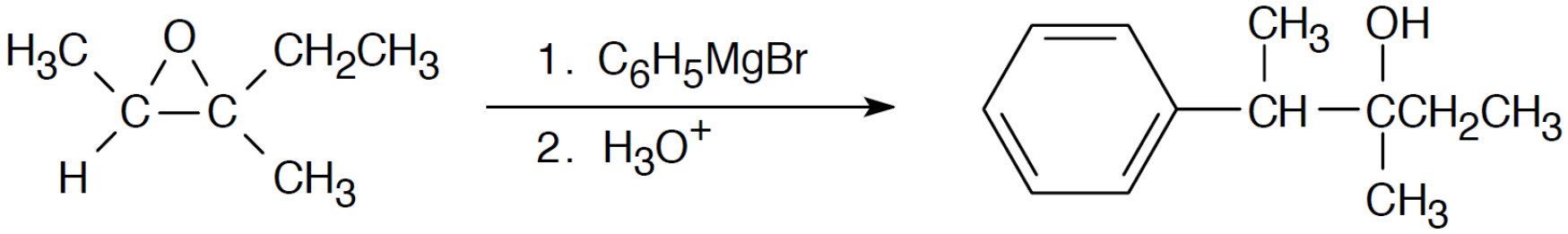
- Solution:

Worked Example

- Predict the major product of the following reaction:



- Solution:



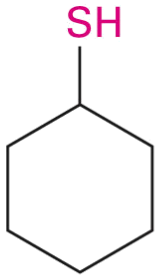
  - Addition of a Grignard reagent takes place at the less substituted epoxide carbon

Thiols and Sulfides

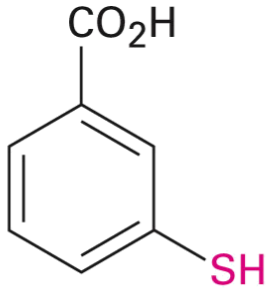
- Thiols
  - Sulfur analogs of alcohols
  - Named with the suffix –thiol
  - –SH group is called **mercapto group**



Ethanethiol



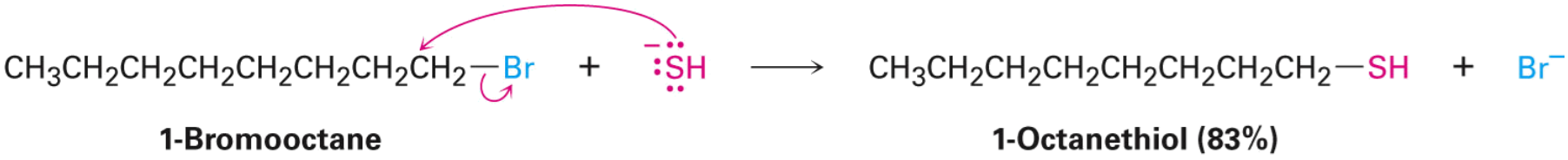
Cyclohexanethiol



***m*-Mercaptobenzoic acid**

Thiols

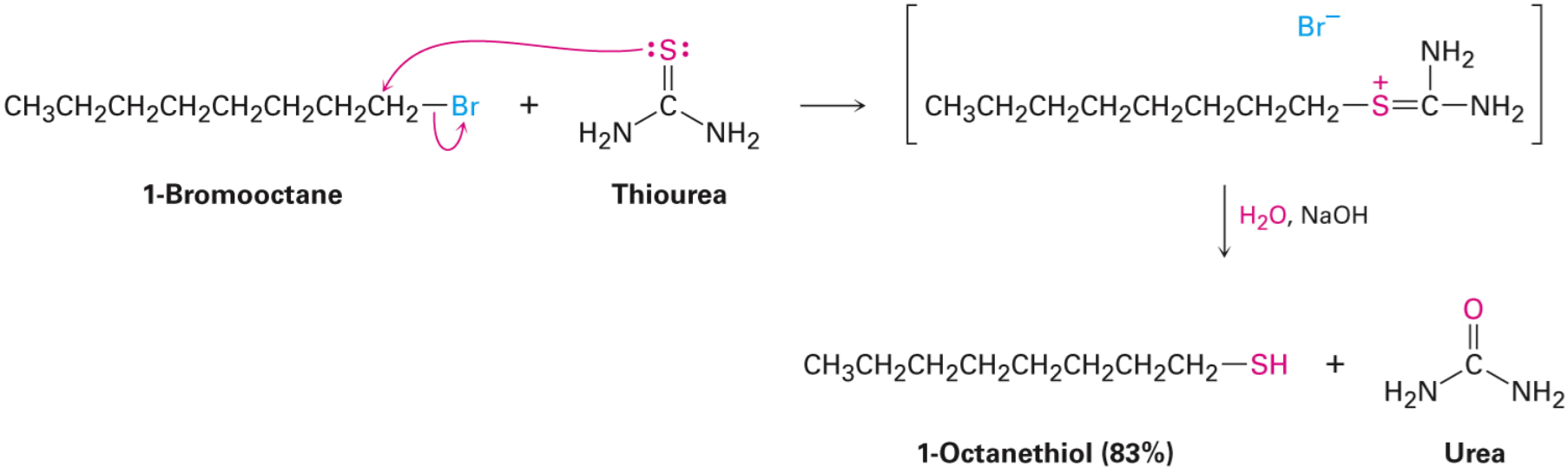
- Prepared from alkyl halides by S_N2 displacement with a sulfur nucleophile



- Alkylthiol product can undergo further reaction with the alkyl halide
  - Gives symmetrical sulfide, a poorer yield of the thiol

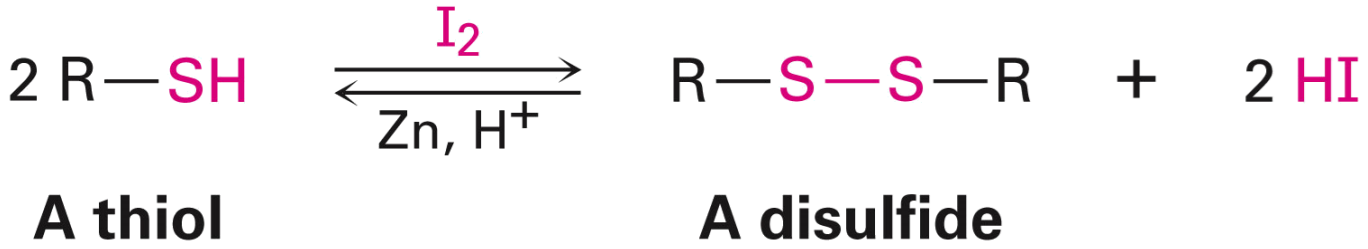
Thiols

- Pure alkylthiol thiourea is used as the nucleophile
  - Gives an intermediate alkyl isothiourea salt, hydrolyzed by subsequent reaction with an aqueous base



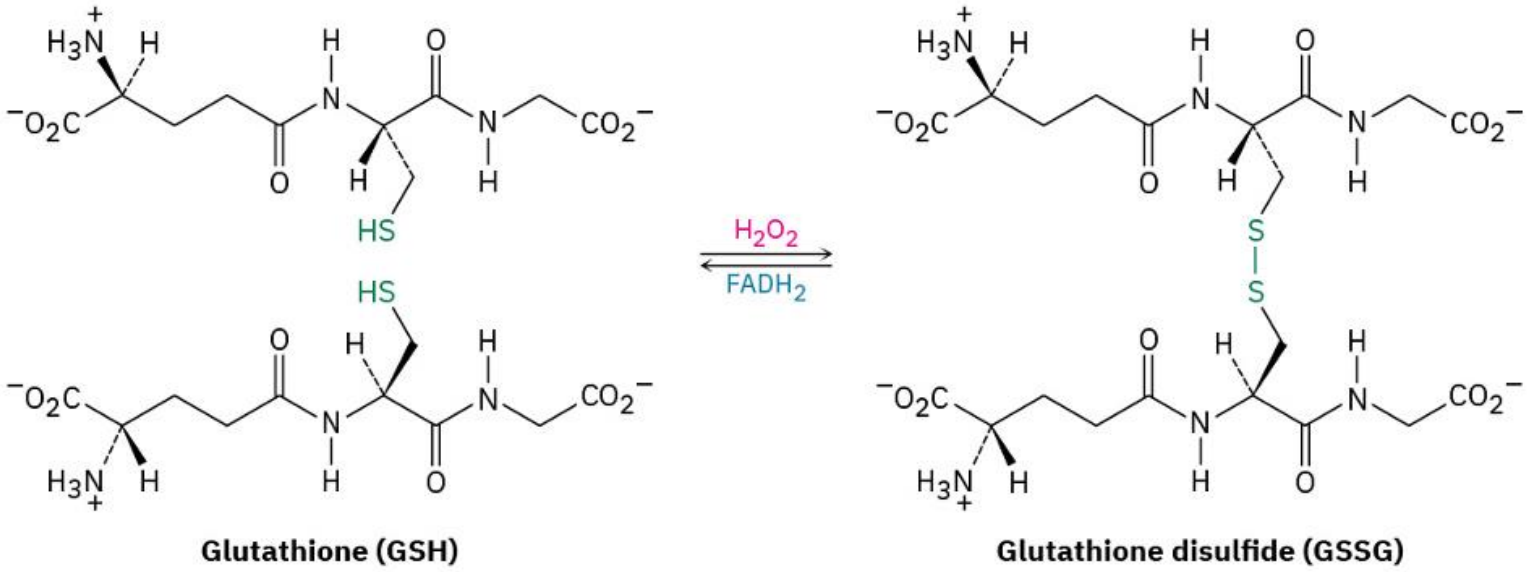
Thiols

- Can be oxidized by Br_2 or I_2
  - Yields **disulfides (RSSR')**



  - Reaction is reversible
  - Key part of numerous biological processes

Thiols



Glutathione (GSH) **Glutathione disulfide (GSSG)**

Reduction back to the thiol requires the coenzyme, reduced $FADH_2$

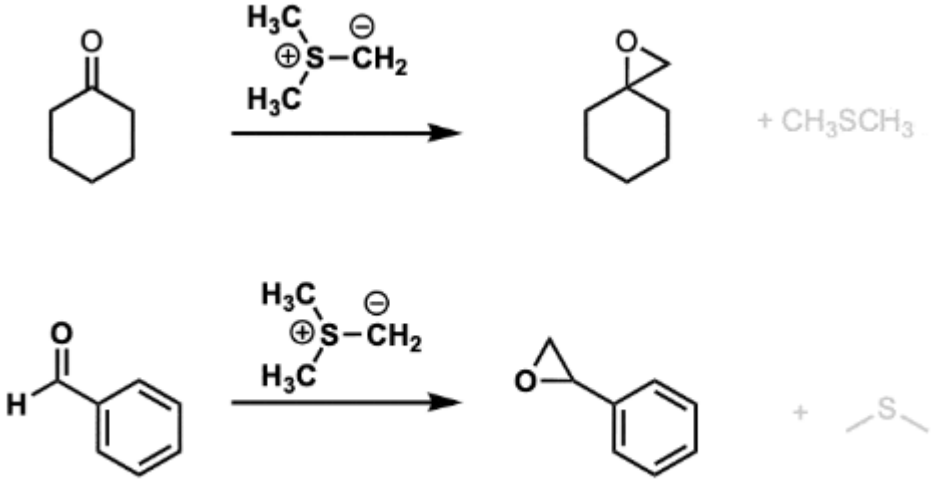
Sulfides

- Thiols can undergo further reaction with the alkyl halide to give a sulfide
- Sulfides and ethers differ substantially in their chemistry
- Through S_N2 mechanism, dialkyl sulfides react rapidly with primary alkyl halides to give **sulfonium ions**

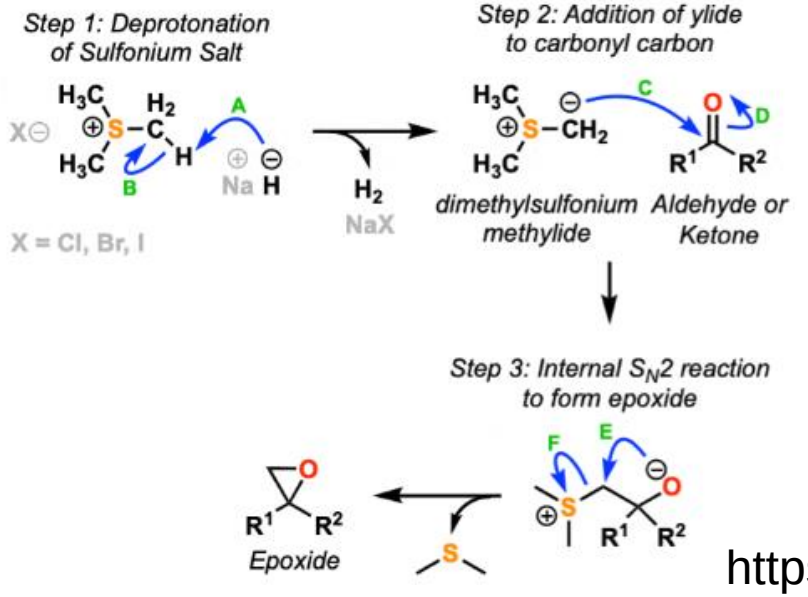


Dimethyl sulfide **Iodomethane** **Trimethylsulfonium iodide**

Johnson–Corey–Chaykovsky reaction



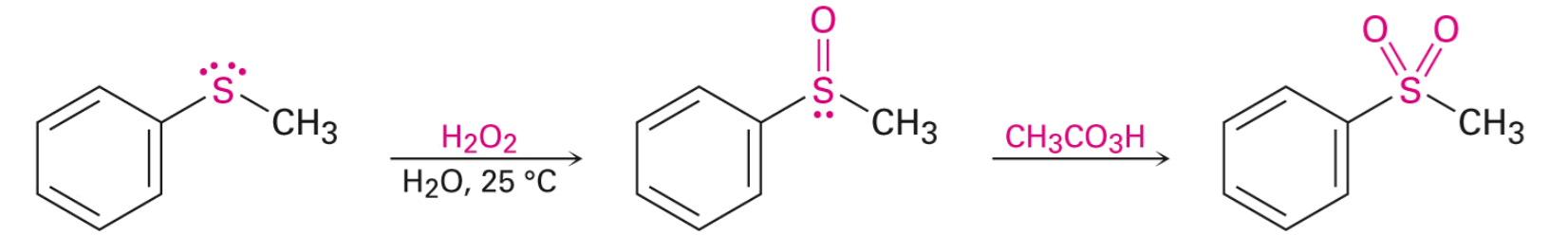
J. Am. Chem. Soc. **1965**, 87(6), 1353-1364



https://www.masterorganicchemistry.com/

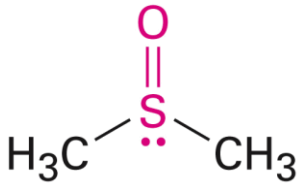
Oxidation of Thiols

- Easily oxidized through treatment of a sulfide with hydrogen peroxide at room temperature
  - Yields **sulfoxide**
  - Further oxidation of the sulfoxide with a peroxyacid yields a **sulfone**



Methyl phenyl sulfide **Methyl phenyl sulfoxide** **Methyl phenyl sulfone**

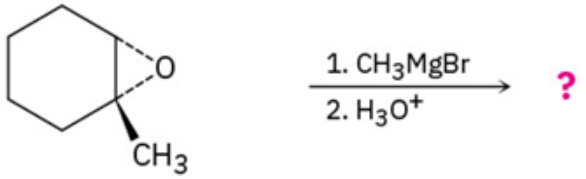
- Dimethyl sulfoxide is often used as a polar aprotic solvent



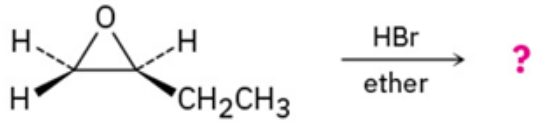
Dimethyl sulfoxide
(a polar aprotic solvent)

PROBLEM 18-27 Predict the product(s) and provide the mechanism for the following reactions:

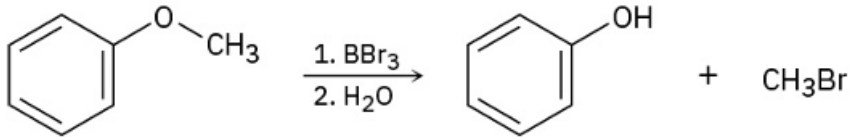
(a)



(b)

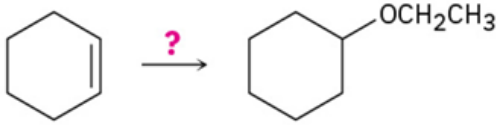


PROBLEM 18-30 Ethers undergo an acid-catalyzed cleavage reaction when treated with the Lewis acid BBr_3 at room temperature. Propose a mechanism for the reaction.

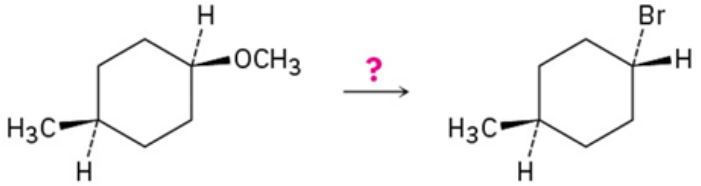


PROBLEM 18-45 How would you carry out the following transformations? More than one step may be required.

(a)



(b)



(c)

